Team Name: Backprop Sust

1. Team Information

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2. Problem Statement

PoliMemeDecode Challenge The challenge is to accurately classify meme images as either "Political" or "Non-Political". This task requires models to understand the subtle intent, satire, and commentary within the meme content, moving beyond mere humor detection. Participants must develop robust models to handle a labeled dataset and predict the correct category for unseen samples. The evaluation criterion is the macro-averaged F1 score, emphasizing balanced performance across both classes. A key difficulty is tackling a distribution-shifted test dataset, necessitating creative preprocessing and modeling strategies for real-world applicability.

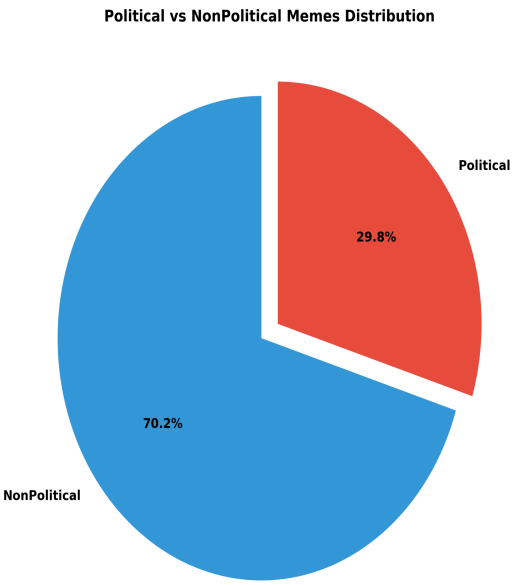
3. Dataset Overview

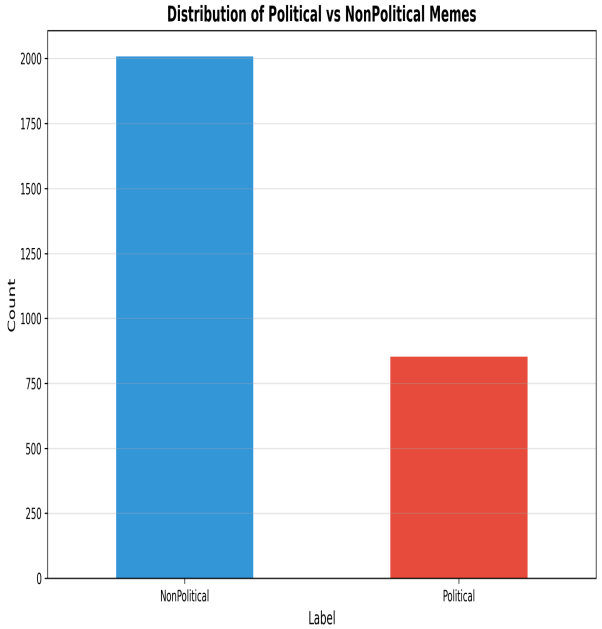
To robustly capture the semantics of political memes, we constructed a dataset by integrating base competition metadata with supplementary textual and visual sources.

- **Primary Data Sources**: We utilized the base training and testing CSVs provided by the competition.The competition features 2860 training images and 330 testing images.
- **Optical Character Recognition (OCR):** We utilized **HunyuanOCR** to extract raw text embedded within the memes. This output was merged with the base training and testing data via unique image identifiers.
- **Synthetic Logo Dataset:** We utilized a background dataset of 20,000 natural images and a collection of clean political logo identifiers. Logos were randomly augmented (position, scale, rotation, flipping) onto backgrounds to simulate the visual diversity and noise of internet memes.

4. Methodology

4.1 Data Visualization

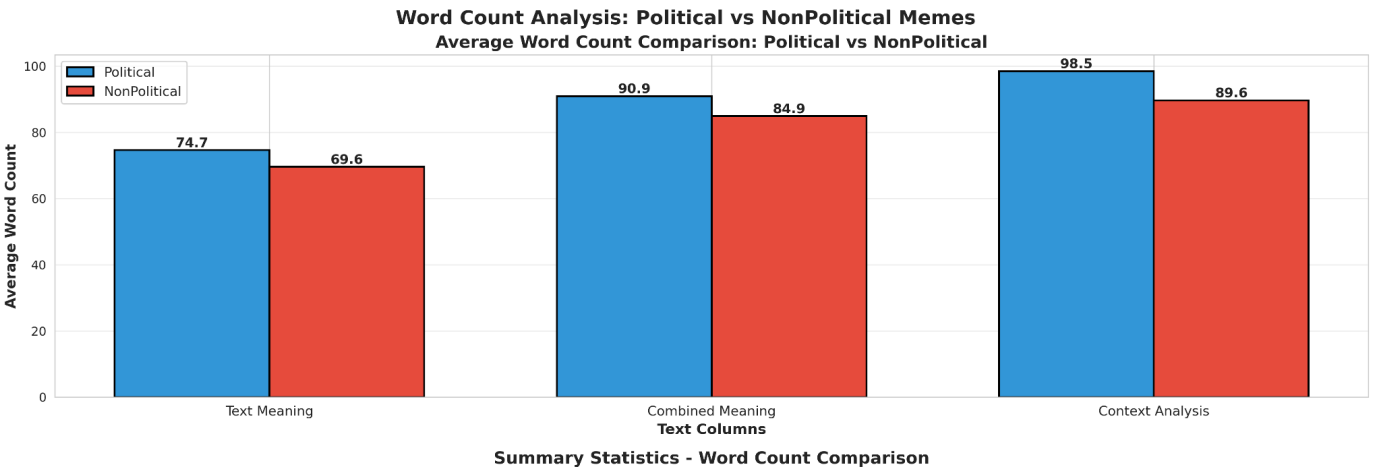




4.2 Data Acquisition and Feature Construction

To robustly capture the semantics of political memes, we treated each sample as a multi-view text object. We constructed the final dataset by integrating the base competition metadata with two supplementary textual sources:

- **Optical Character Recognition (OCR):** We utilized HunyuanOCR to extract raw text embedded within the memes, merging this output with the base training and testing CSVs via unique image identifiers. This feature is denoted as `text`.
- **Semantic Enrichment (Gemma Multimodal Inference):** To address the subtlety of political satire, we employed the Gemma model to generate rich semantic descriptions. We fed both the original meme image and the extracted OCR text into the model to ground the generation in both visual and textual cues. From the model's output, we extracted three distinct high-level features:
    - `text_meaning`: An interpretation of the literal text within the meme.
    - `combined_meaning`: A synthesis of the visual and textual elements representing the meme's surface message.
    - `context_analysis`: A deeper explanation of the underlying background and specific political references.



Column	Political Mean	NonPolitical Mean	Difference	% Diff
Text Meaning	74.7	69.6	+5.1	+7.3%
Combined Meaning	90.9	84.9	+6.0	+7.1%
Context Analysis	98.5	89.6	+8.8	+9.9%

4.3 Model Selection

We adopted a multi-stage pipeline combining a Dual-Encoder Transformer for text analysis and visual heuristics (YOLOv8 + InsightFace) for image analysis.

A. Dual-Encoder Text Classifier

We designed a custom architecture that leverages two distinct pre-trained backbones to process the multi-view text data:

- **Primary Encoder:** `microsoft/deberta-v3-large`
- **Secondary Encoder:** `RoBERTa-Large`

Feature Fusion Mechanism: To resolve ambiguities (e.g., sarcasm), the features interact via a **Multi-Head Cross-Attention** mechanism (8 heads). The *Combined Meaning* stream attends to the *Context Analysis* stream, and vice versa.

The processed features are concatenated into a unified representation and passed through a **Gated Fusion** mechanism to dynamically filter information

The final classification is performed by a 3-layer MLP (1024 → 512 → 256) with GELU activation and LayerNorm.

B. Visual Heuristics (Post-Processing Pipeline)

To capture visual cues that text models miss, we implemented a cascaded pipeline:

1. **Logo Recognition (YOLOv8):** We fine-tuned a YOLOv8-Nano model on our synthetic dataset. During inference, if a political logo is detected with confidence >0.88, the prediction is overridden to "Political".
2. **Face Recognition (InsightFace):** We constructed a reference gallery of embeddings (using the `buffalo_l` model) for prominent politicians in Bangladesh and India. If a test image matches a gallery entry with Cosine Similarity >0.5, it is flagged as "Political".
3. **Keyword Filtering:** We scanned the text for a compiled regex of political entities (e.g., "Awami League", "Modi") and terms in Bengali and English.

C. Training Strategy

- **Validation:** 5-Fold Stratified Cross-Validation.
- **Optimization:** AdamW optimizer with Cosine Annealing Warm Restarts.
- **Regularization:** Dropout (0.2–0.5) and Gradient Clipping (norm 1.0).
- **Inference:** Soft Voting ensemble across all 5 folds.

5. Results and Analysis

Table 1: Performance Comparison of Proposed Methods (Macro F1 Score)

Method	Configuration	Public LB	Private LB
Baseline	Text Only (**DeBERTa-v3-Large + RoBERTa-Large**)	0.83524	0.84061
Method A	Text + Logo Detection (YOLOv8)	0.84391	0.84539
Method B (Final)	Text + Logo + Face Recognition (InsightFace)	0.83923	**0.84622**

References

Our solution, which utilizes multimodal feature fusion, MLLM-based semantic enrichment, and dual-stage classification, is directly inspired by recent advances in multimodal meme analysis, specifically:

[1] Shi, Y., Meng, W., Guo, Z., Wei, C., & Chen, W. (2025). *Enhancing Meme Emotion Understanding with Multi-Level Modality Enhancement and Dual-Stage Modal Fusion*. arXiv preprint arXiv:2511.11126. Available at: https://arxiv.org/abs/2511.11126